Tutorial-5

An email service provider wants to classify emails spam or not spam using linear svm classifier

Input Features:

- no. of suspicious words
- no. of external links
- Attachment present (0/1)

Target varibles

class (spam=1, notspam=-1)

suspicious words (x_1)	links (x_2)	Attachment (x_3)	class (y)
6	3	1	-1
1	0	0	-1
4	2	1	+1
0	0	0	-1
5	2		+1

Initial weights $w_1 = w_2 = w_3 = 1$, $b = 0$, $\eta = 1$

sample 1: $(6, 3, 1)$ $\boxed{y = 1}$

$$\boxed{f(x) = w_1 x_1 + w_2 x_2 + w_3 x_3 + b}$$

$$f(x) = 6(1) + 3(1) + 1(1) + 0$$

$$= 10$$

$$y \cdot (f(x)) = y(f(x)) = (-1)(10) = -10$$

If $y[f(x)] \geq 1 \rightarrow$ Perfect classification

If $y[f(x)] < 1 \rightarrow$ miss classification

Here $y(f(x)) < 1$, so miss calculation.

Updating weights and bias:

$$w_0(new) = w_0(old) + \eta \sum_{i=1}^{n} c_i y_i \quad \boxed{c=1}$$

$$w_0 new = 0 + 1((1)(-1)] \Rightarrow -1$$

$$w_1 new = w_1(old) + -\eta(w_1(old) - \sum_{i=1}^{n} (i\, y_i\, x_i]$$

$$= 1 - 1[1 - 1(-1)(6)] \Rightarrow -6$$

$$w_2 new = 1 - 1[1 - (1)(-1)(3)] \Rightarrow -3$$

$$w_3 new = 1 - 1[1 - 1(1)(1)] \Rightarrow -1$$

sample ②: (1,0,0) $\boxed{y=-1}$

$f(x) = (-6)(1) + (-3)(0) + (-1)(0) + (-1) = -7$

$y(f(x)) = (-1)(-7) = 7$

$y(f(x)) \geq 1$ perfect classification

updating weights

$w_{1(new)} = (-6) - (1)(-6) = 0$

$w_{2(new)} = (-3) - (1)(-3) = 0$

$w_{3(new)} = (-1) - (1)(-1) = 0$

$\boxed{w_{0(new)} = w_{0(old)}}$

sample ③: (4,2,1) $\boxed{y=+1}$ $(w_1, w_2, w_3, w_0) = (0,0,0,-1)$

$f(x) = 0\times4 + 0(2) + 0(1) + (-1) = -1$

$y(f(x)) = (-1)(1) = 1$

$y(f(x)) < 1 \rightarrow$ miss classification

updating weights and bias

$w_{1(new)} = 0 - 1(0 - 1(1)(4)) = 4$

$w_{2(new)} = 0 - 1[0 - 1(1)(2)] = 2$

$w_{3(new)} = 0 - 1[0 - 1(1)1] = 1$

$w_{0(new)} = -1 + 1(1) = 0$

sample ④: (0,0,0) $\boxed{y=-1}$ $(w_1, w_2, w_3, w_0) = (4,2,1,0)$

$f(x) = 4(0) + 2(0) + 1(0) + 0 = 0$

$y[f(x)] = -1\times0 = 0$

$y[f(x)] < 1 \rightarrow$ miss classification

updating weight and bias

$w_{1(new)} = 4 - 1[4 - 1(-1)0] = 0$

$w_{2(new)} = 2 - 1[2 - 1(-1)0] = 0$

$w_{3(new)} = 1 - 1[1 - 1(-1)0] = 0$

$w_{0(new)} = 0 + 1(+1) = -1$

Sample = St(5, 2, 1) $(w_0, w_1, w_2, w_3) = (-1, 0, 0, 0)$

$f(x) = 0(5) + 0(2) + 0(1) + (-1) = -1$

$y(f(x)) \neq 1$ - missclassification

updating weights and bias

$w_1(new) = 0 - 1[0 - 1(1)5] = 5$

$w_2(new) = 0 - 1[0 - 1(1)2] = 2$

$w_3(new) = 0 - 1[0 - 1(1)1] = 1$

$w_0(new) = -1 + 1(1)(1) = 0$

Final weight and bias are:

$w_1 = 5, \; w_2 = 2, \; w_3 = 1, \; w_0 = 0$

$$\text{margin} = \frac{2}{\|w\|} = \frac{2}{\sqrt{w_1^2 + w_2^2 + w_3^2}}$$

$$= \frac{2}{\sqrt{(5)^2 + (2)^2 + (1)^2}} = \frac{2}{\sqrt{30}} \simeq 0.365$$